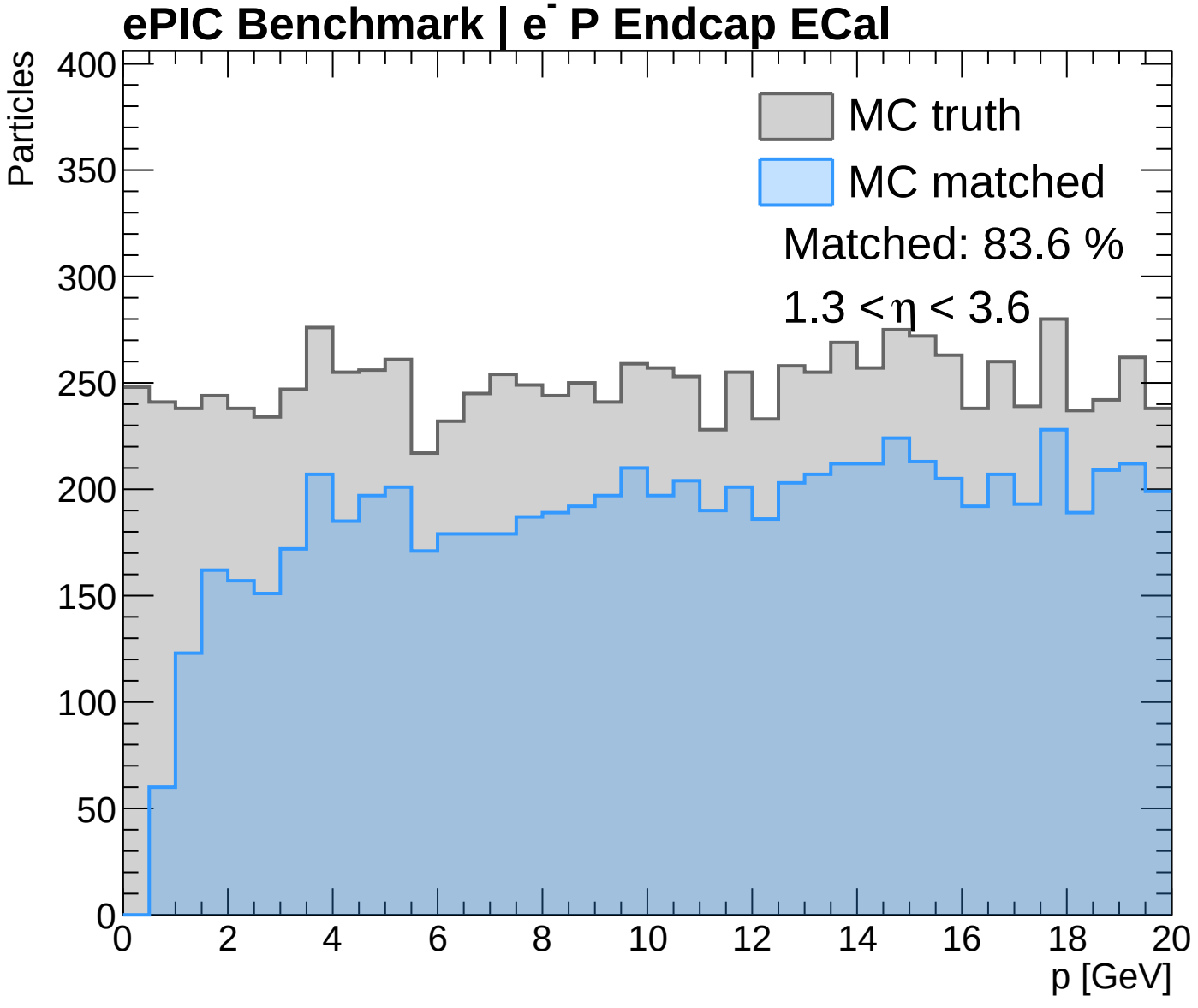
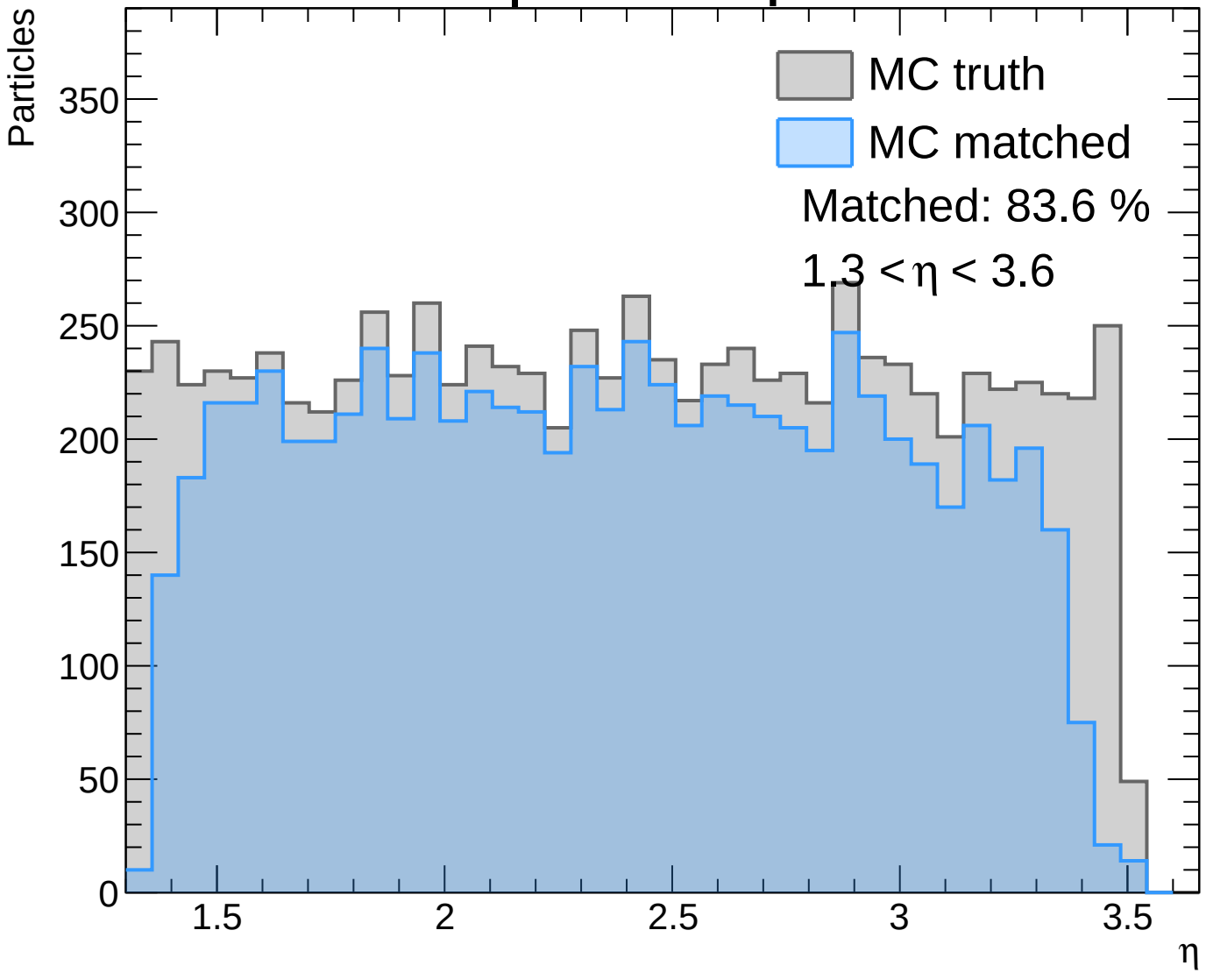
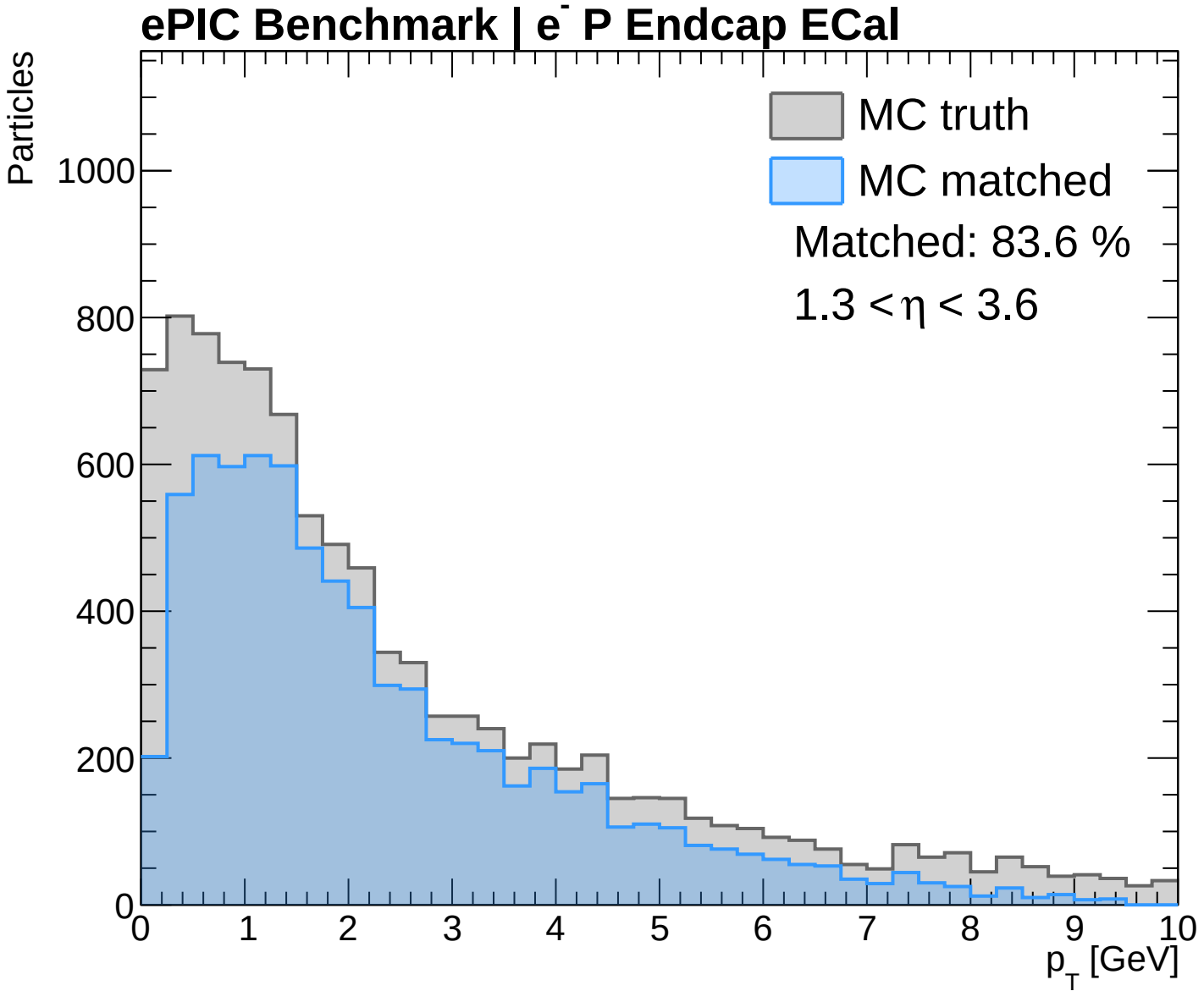
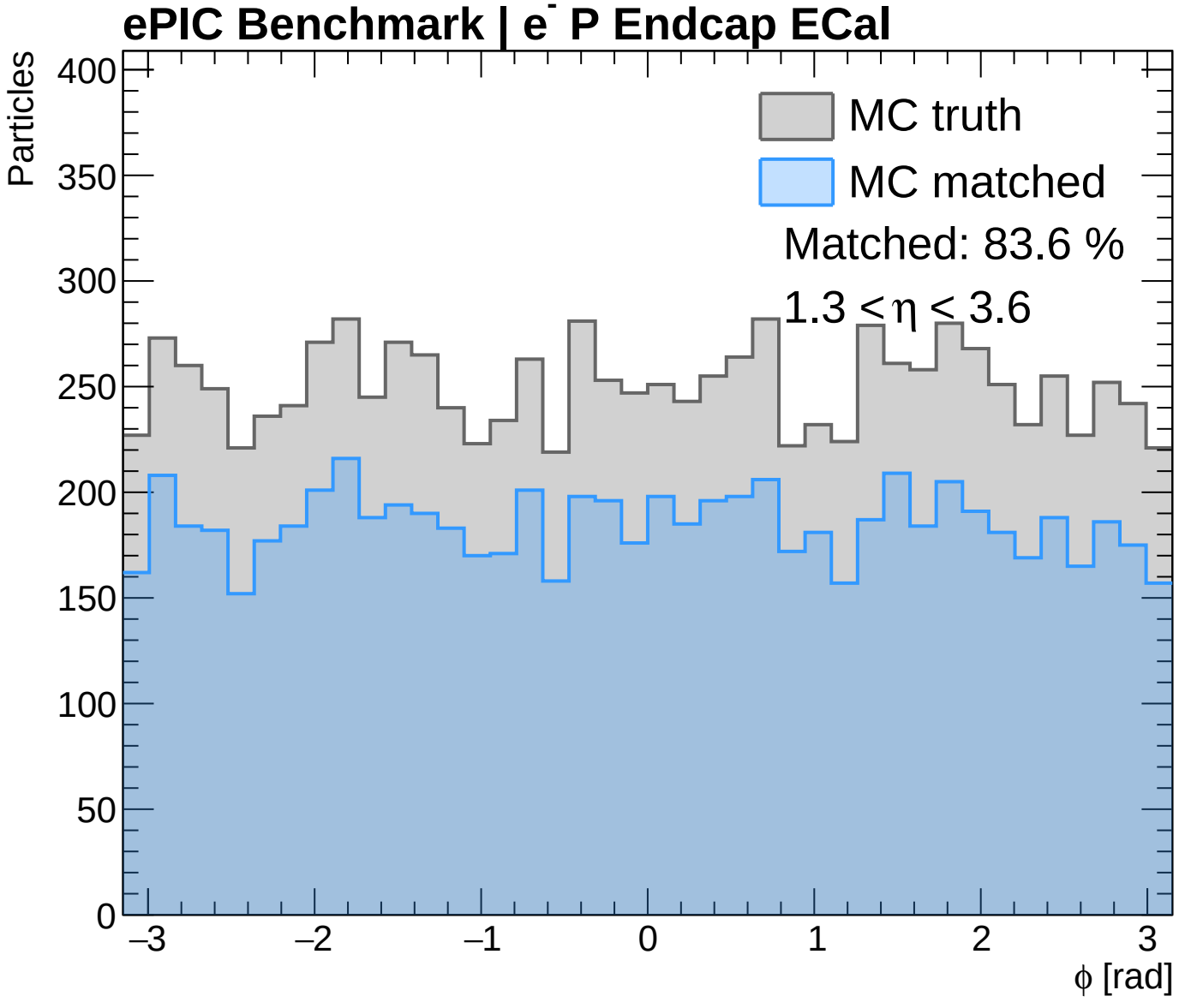
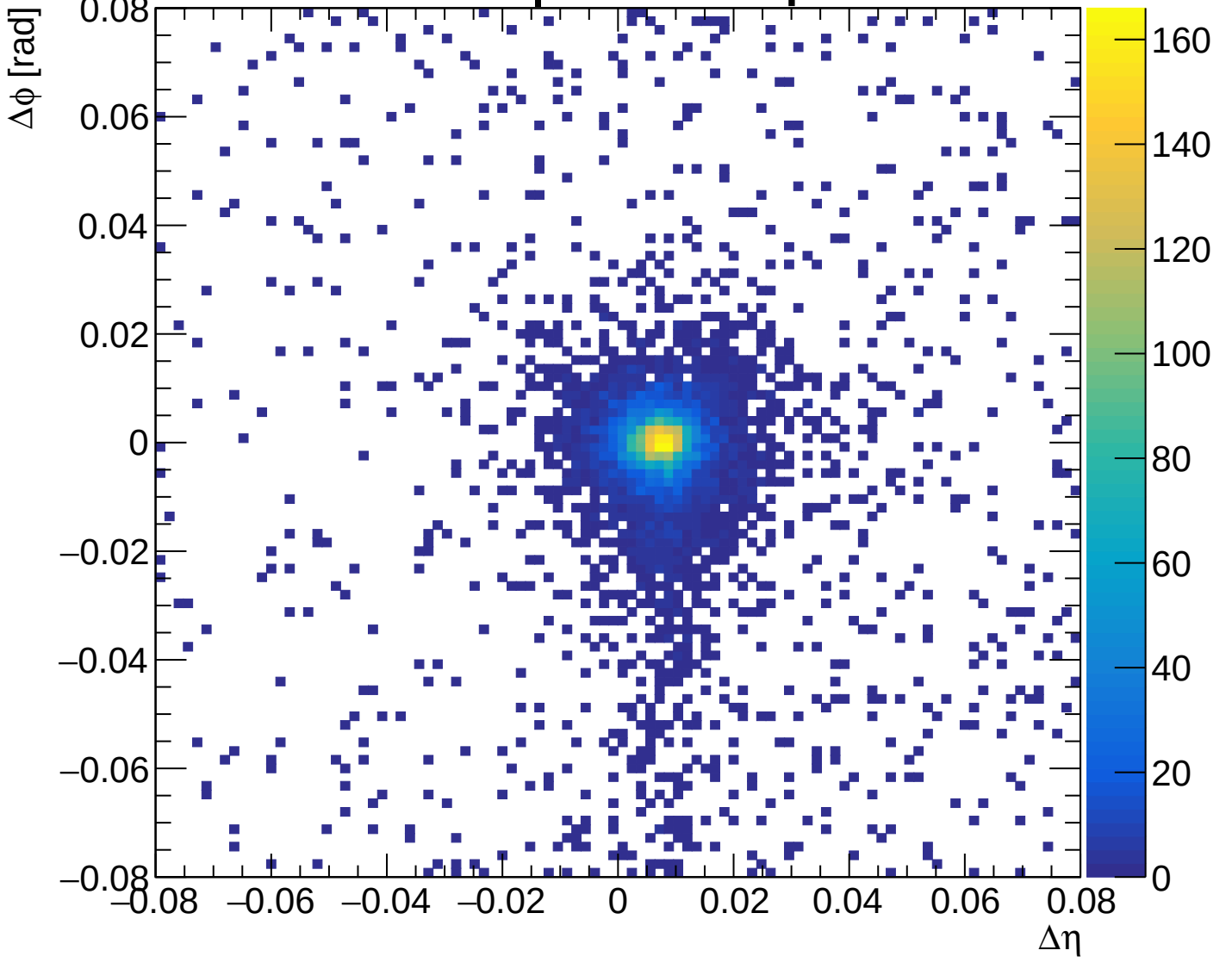
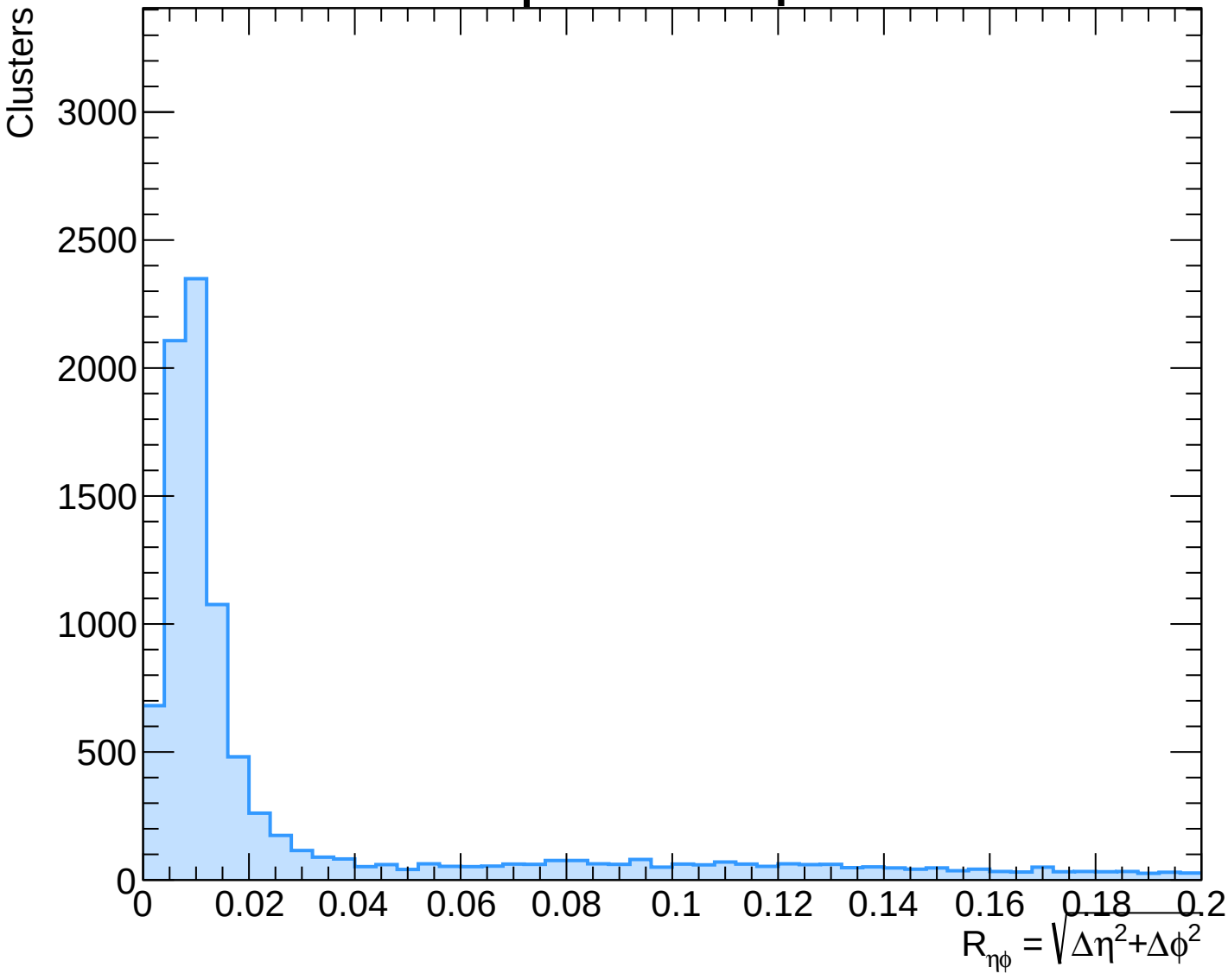

ePIC Benchmark | e- P Endcap ECal
Particles
400
350
300
250
200
150
100
50
0
0
2
4
6
8
10
12
14
16
18
20
p [GeV]
MC truth
MC matched
Matched: 83.6 %
1.3 < η < 3.6

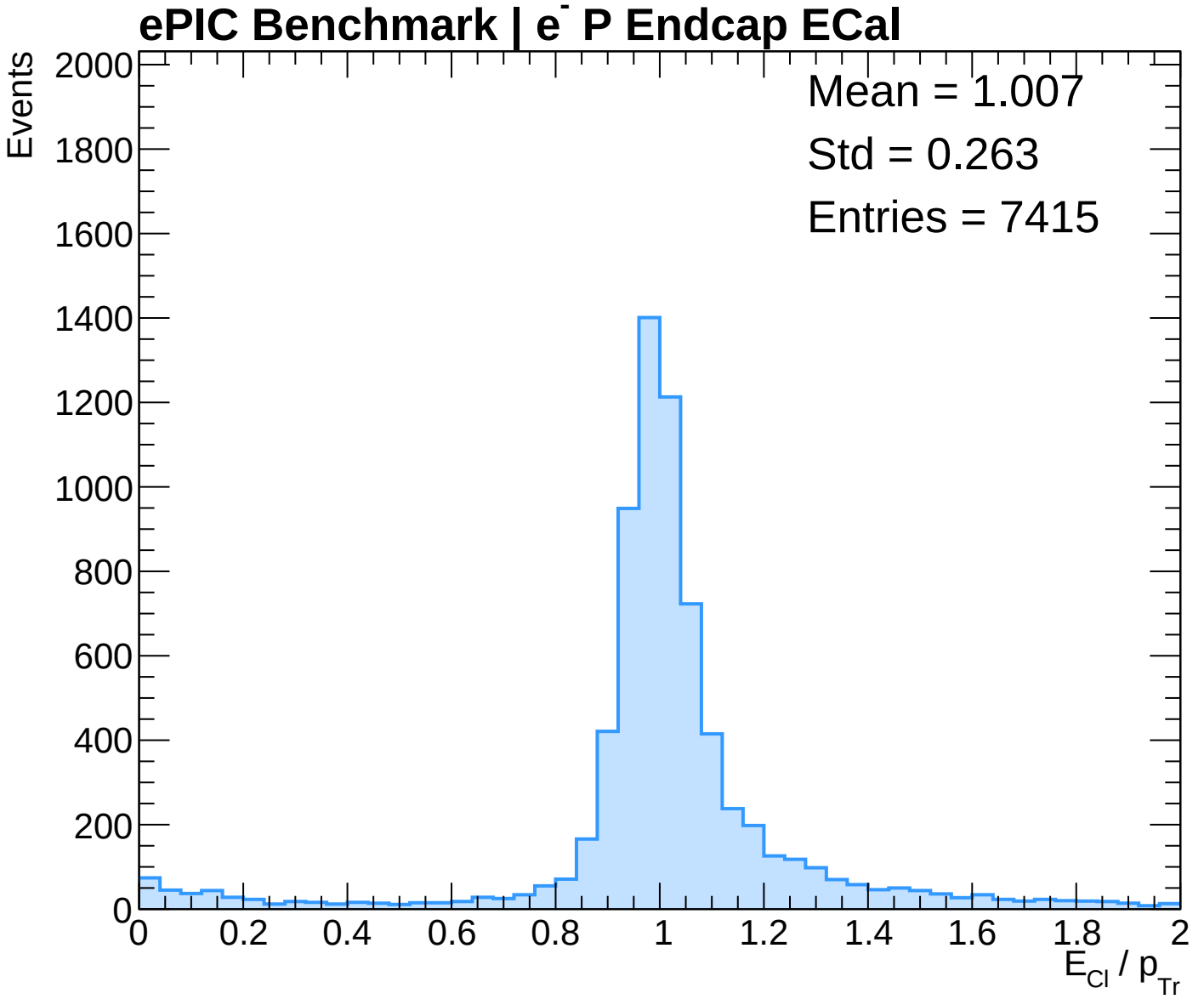

ePIC Benchmark | e⁻ P Endcap ECal
Particles
MC truth
MC matched
Matched: 83.6 %
1.3 < η < 3.6
350
300
250
200
150
100
50
0
1.5
2
2.5
3
3.5
η

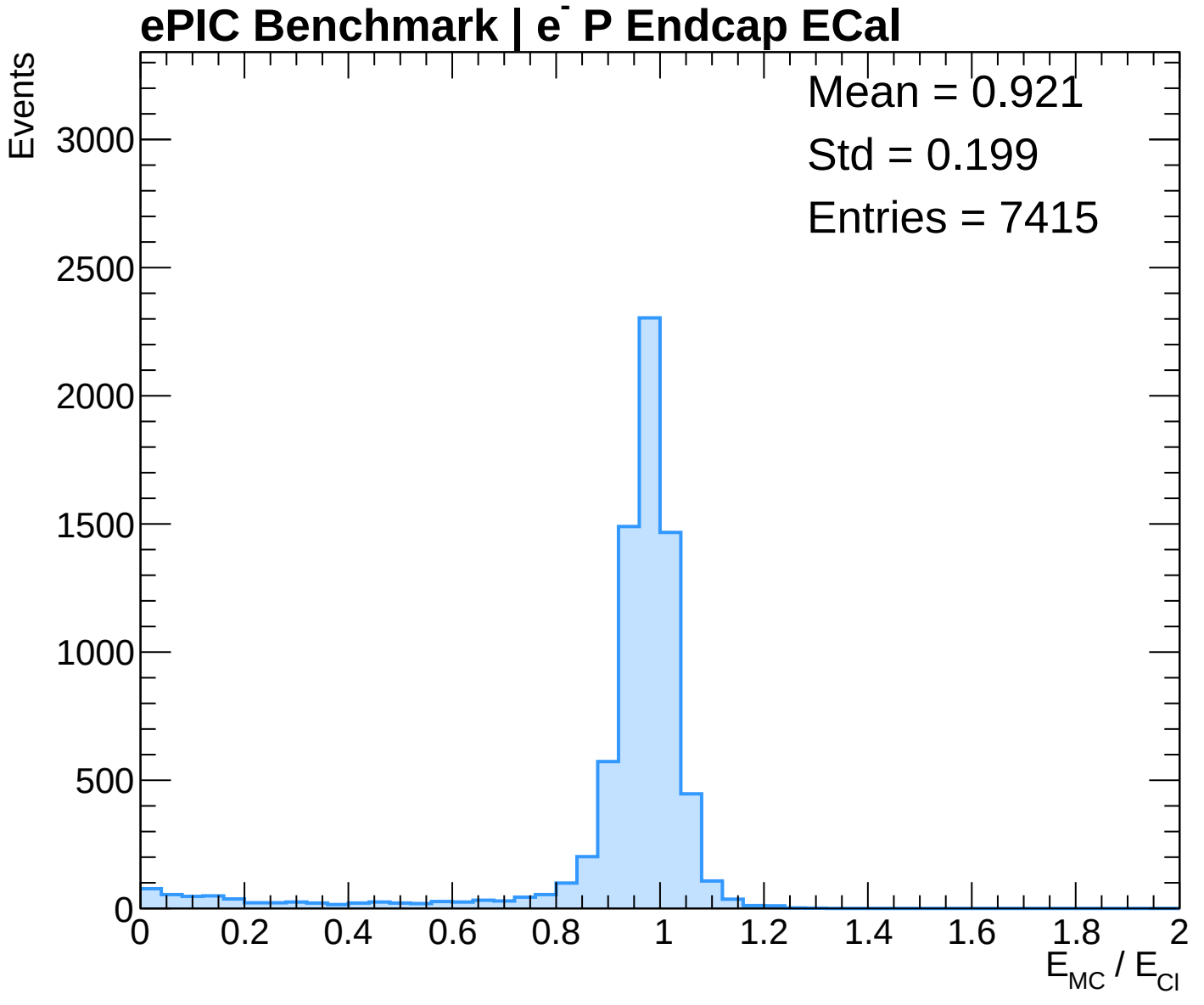

ePIC Benchmark | e- P Endcap ECal
Particles
MC truth
MC matched
Matched: 83.6 %
1.3 < η < 3.6
0
200
400
600
800
1000
0
1
2
3
4
5
6
7
8
9
10
pT [GeV]

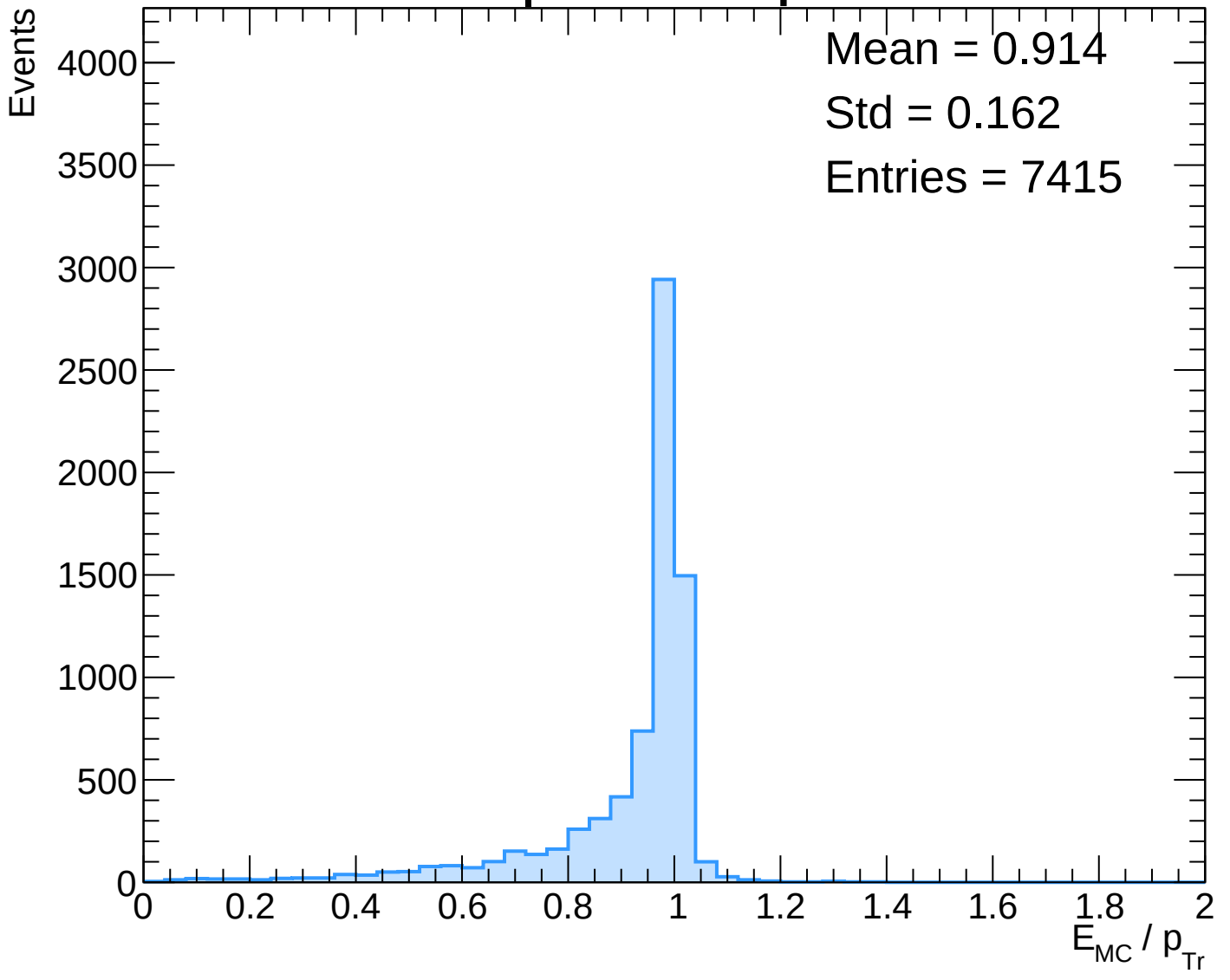

ePIC Benchmark | e- P Endcap ECal
Particles
400
350
300
250
200
150
100
50
0
–3
–2
–1
0
1
2
3
ϕ [rad]
MC truth
MC matched
Matched: 83.6 %
1.3 < η < 3.6

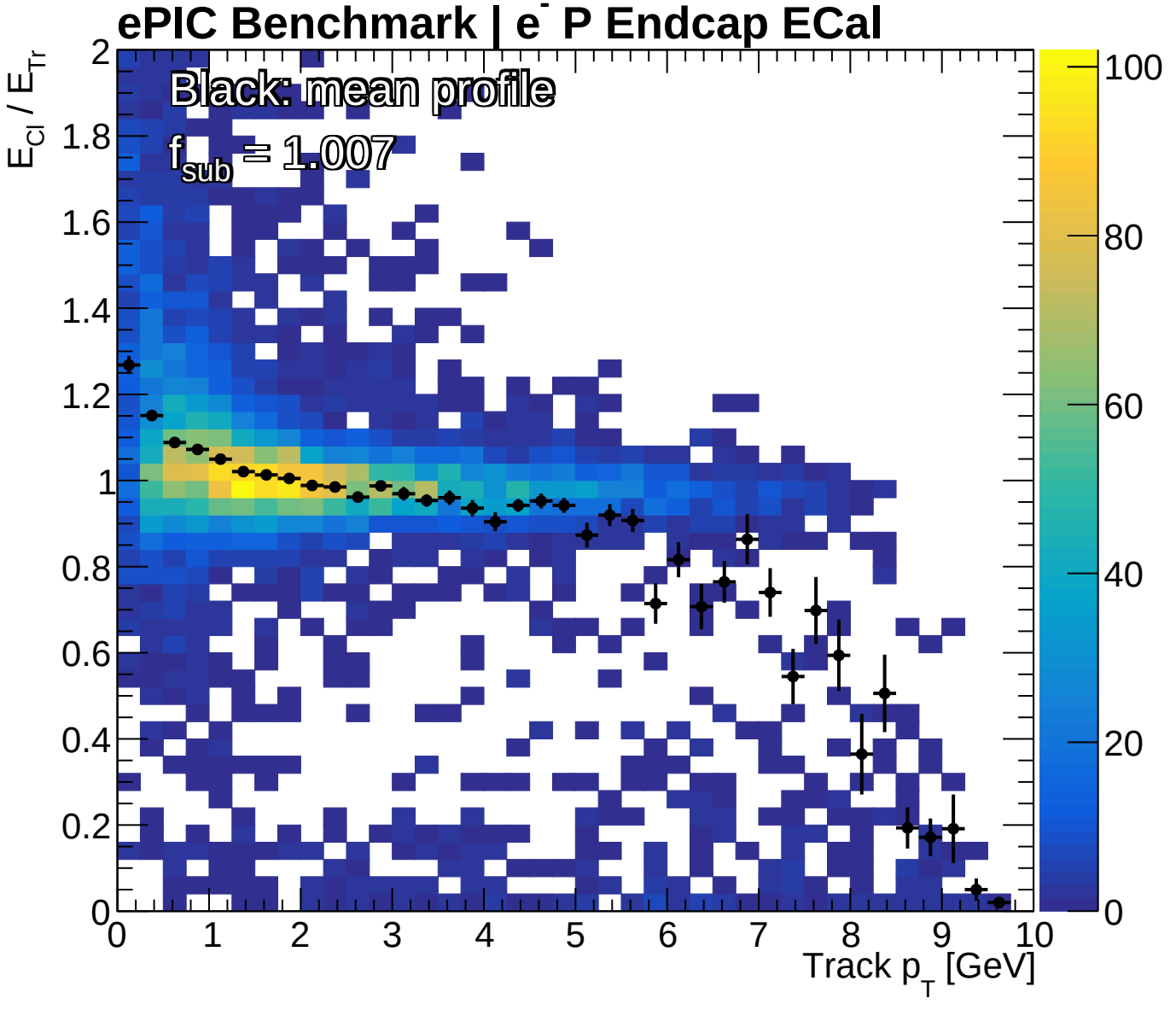

ePIC Benchmark | e⁻ P Endcap ECal
Δϕ [rad]
0.08
0.06
0.04
0.02
0
−0.02
−0.04
−0.06
−0.08
−0.08
−0.06
−0.04
−0.02
0
0.02
0.04
0.06
0.08
Δη
160
140
120
100
80
60
40
20
0

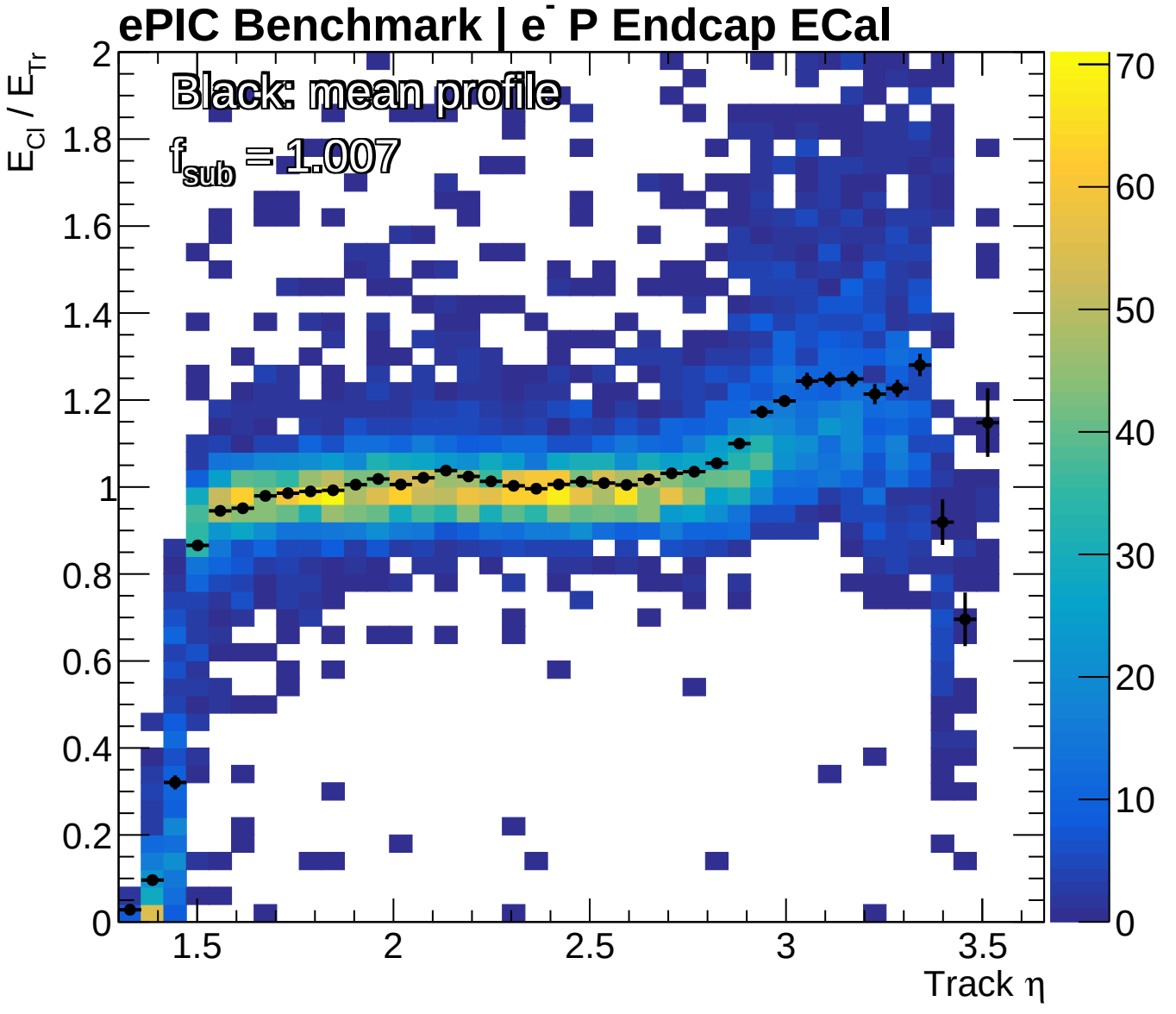

ePIC Benchmark | e- P Endcap ECal
Clusters
3000
2500
2000
1500
1000
500
0
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
$R_{\eta\phi} = \sqrt{\Delta\eta^2+\Delta\phi^2}$

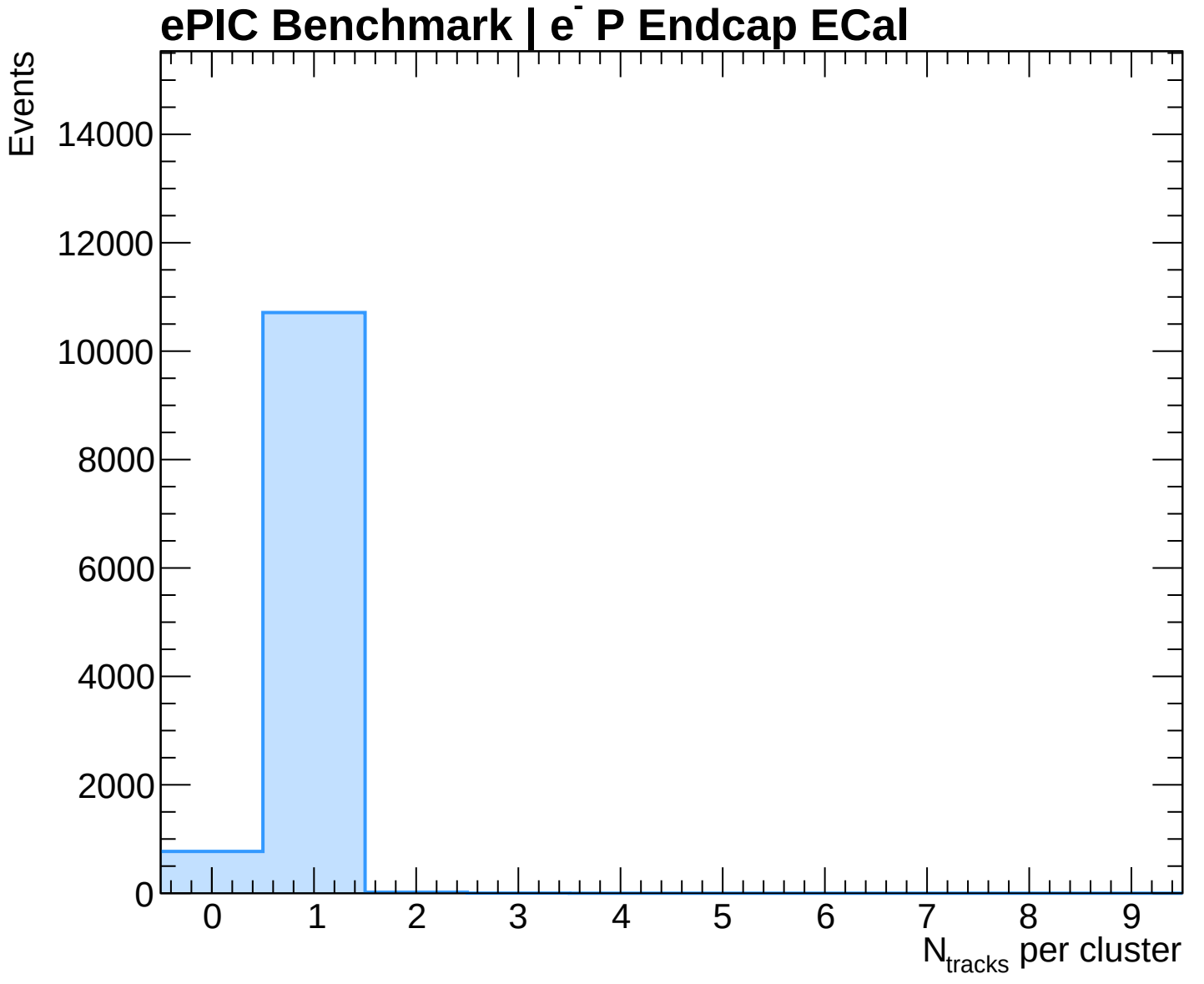

ePIC Benchmark | e- P Endcap ECal
Mean = 1.007
Std = 0.263
Entries = 7415
Events
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_Cl / p_Tr

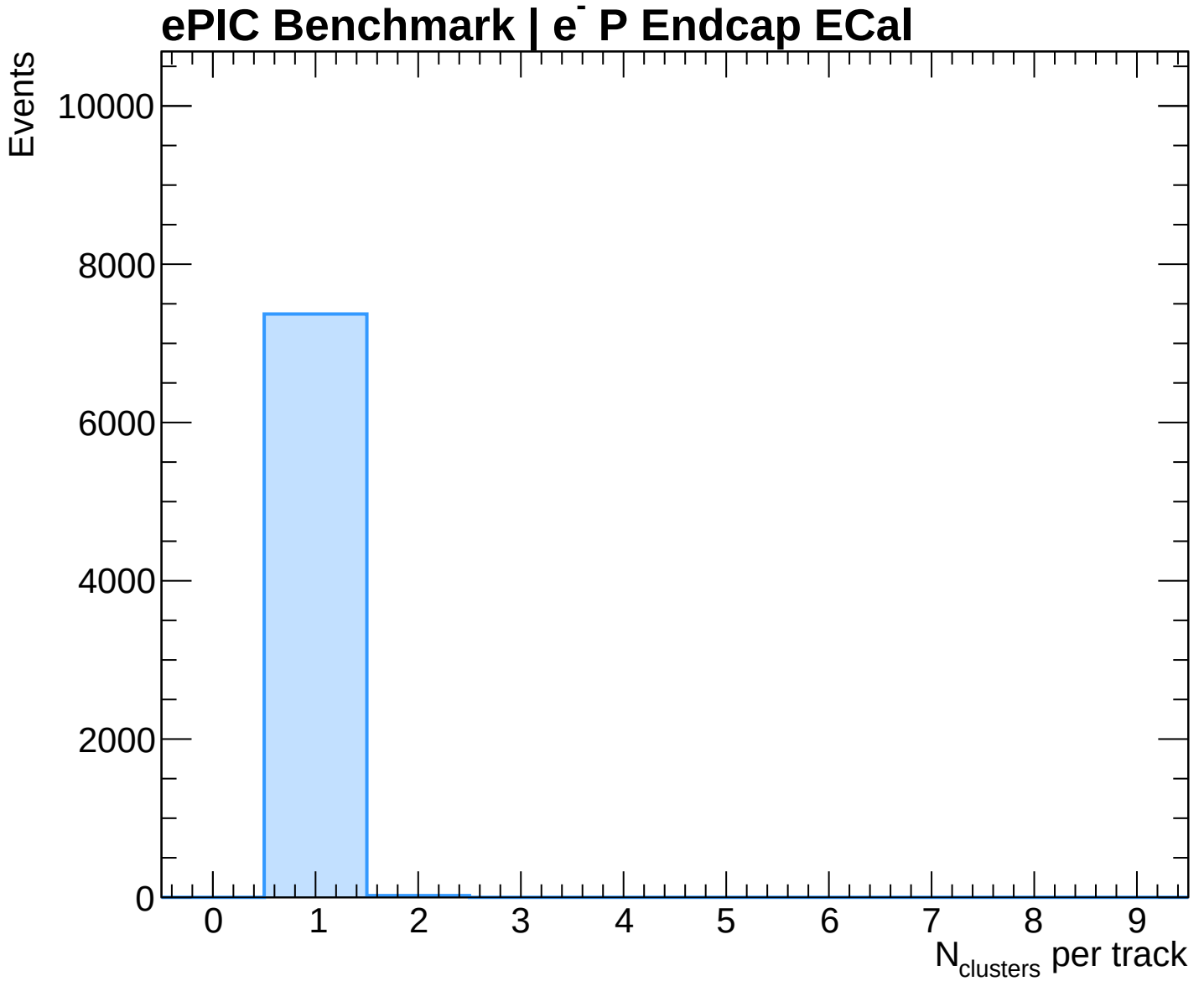

ePIC Benchmark | e- P Endcap ECal
Mean = 0.921
Std = 0.199
Entries = 7415
Events
3000
2500
2000
1500
1000
500
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_MC / E_Cl

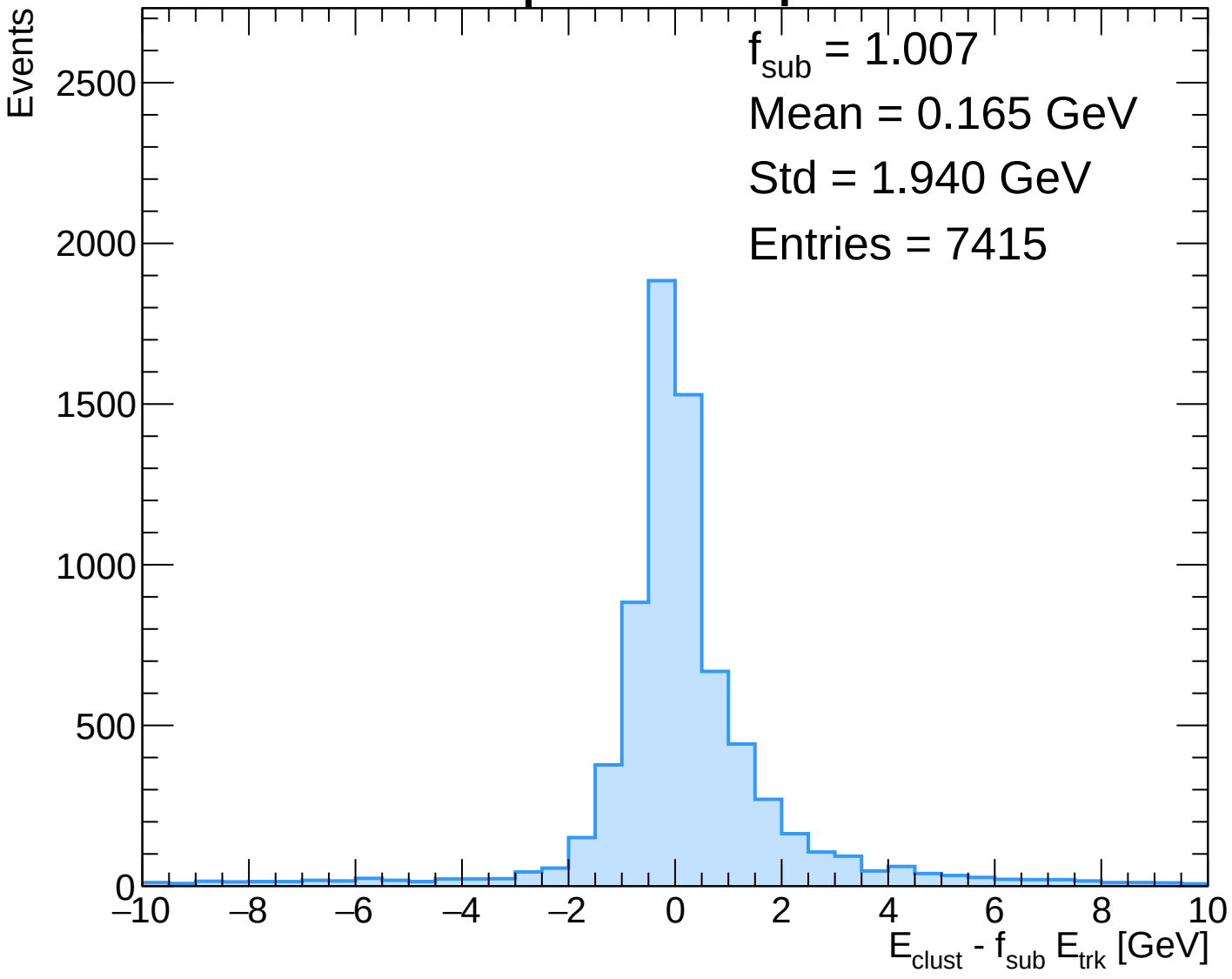

ePIC Benchmark | e- P Endcap ECal
Mean = 0.914
Std = 0.162
Entries = 7415
Events
4000
3500
3000
2500
2000
1500
1000
500
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_MC / p_Tr


ePIC Benchmark | e⁻ P Endcap ECal
Black: mean profile
f_sub = 1.007
E_Cl / E_Tr
Track p_T [GeV]
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0
1
2
3
4
5
6
7
8
9
10
0
20
40
60
80
100


ePIC Benchmark | e⁻ P Endcap ECal
Black: mean profile
f_{sub} = 1.007
E_{Cl} / E_{Tr}
Track η
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
1.5
2
2.5
3
3.5
0
10
20
30
40
50
60
70


ePIC Benchmark | e- P Endcap ECal
Events
14000
12000
10000
8000
6000
4000
2000
0
0
1
2
3
4
5
6
7
8
9
N_tracks per cluster


ePIC Benchmark | e- P Endcap ECal
Events
10000
8000
6000
4000
2000
0
0
1
2
3
4
5
6
7
8
9
N clusters per track


ePIC Benchmark | e- P Endcap ECal
Events
2500
2000
1500
1000
500
0
f_sub = 1.007
Mean = 0.165 GeV
Std = 1.940 GeV
Entries = 7415
–10
–8
–6
–4
–2
0
2
4
6
8
10
E_clust - f_sub E_trk [GeV]